T1 Because of $\theta(u,v) = \inf\{f(x) + u^T g(x) + v^T h(x) : x \in X\}$

$\lambda \in [0,1]$

$\therefore \theta(\lambda u_1 + (1-\lambda)u_2, \lambda v_1 + (1-\lambda)v_2)$

$= \inf\{f(x) + (\lambda u_1 + (1-\lambda)u_2)^T g(x) + (\lambda v_1 + (1-\lambda)v_2)^T h(x) : x \in X\}$

$= \inf\{f(x) + \lambda u_1^T g(x) + (1-\lambda)u_2^T g(x) + \lambda v_1^T h(x) + (1-\lambda)v_2^T h(x) : x \in X\}$

$\theta(u_1, v_1) = \inf\{f(x) + u_1^T g(x) + v_1^T h(x) : x \in X\}$

$\theta(u_2, v_2) = \inf\{f(x) + u_2^T g(x) + v_2^T h(x) : x \in X\}$

$\therefore \lambda\theta(u_1, v_1) + (1-\lambda)\theta(u_2, v_2) = \lambda \inf\{f(x) + u_1^T g(x) + v_1^T h(x) : x \in X\} + (1-\lambda)\inf\{f(x) + u_2^T g(x) + v_2^T h(x) : x \in X\}$

$= \inf\{\lambda f(x) + \lambda u_1^T g(x) + \lambda v_1^T h(x) : x \in X\} + \inf\{(1-\lambda)f(x) + (1-\lambda)u_2^T g(x) + (1-\lambda)v_2^T h(x) : x \in X\}$

$\leq \inf\{\lambda f(x) + (1-\lambda)f(x) + \lambda u_1^T g(x) + (1-\lambda)u_2^T g(x) + \lambda v_1^T h(x) + (1-\lambda)v_2^T h(x) : x \in X\}$

$= \theta(\lambda u_1 + (1-\lambda)u_2, \lambda v_1 + (1-\lambda)v_2)$

$\therefore$ The Lagrangian dual function is convex function.

T2 (a) According to the title, we can set an IP for this case:

minimize $\sum_J f_j \cdot z_j + \sum_I \sum_J x_{ij} \cdot d_{ij} \cdot \alpha$

s.t.:

demand limit: $h_i \leq \sum_J x_{ij}$, $\forall i \in I$

capacity limit: $z_j k_j \geq \sum_I x_{ij}$, $\forall j \in J$

x_{ij} represents the amount of delivery from facility J to customer node I. (integer)

$z_j = \begin{cases} 1, & \text{if facility is build in place } J \\ 0, & \text{else.} \end{cases}$

(b) if a customer can only be served by one facility

Assume that $y_{ij} = \begin{cases} 1, & \text{facility } J \text{ serves customer } I. \\ 0, & \text{else.} \end{cases}$

minimize $\sum_J f_j \cdot z_j + \alpha \sum_I \sum_J x_{ij} \cdot y_{ij} \cdot d_{ij}$

s.t.

demand limit: $h_i \leq \sum_J x_{ij} \cdot y_{ij}$

capacity limit: $z_j k_j \geq \sum_I x_{ij} y_{ij}$

a customer can only served by one facility: $\sum_J y_{ij} = 1$

if $y_{ij}=0$, $x_{ij}=0$: $\qquad M y_{ij} \geq x_{ij}$

x_{ij} represents the amount of delivery from facility J to customer node I. (integer)

$z_j = \begin{cases} 1 & \text{, if facility is build in place J} \\ 0 & \text{, else.} \end{cases}$